This is for metric space course.

1.1 Metric Space

Definition 1.1.1 (Metric space)**.** Let X be a non-empty set. A metric is a function $d : X \times X \to \mathbb{R}$, such that for all $x, y, z \in X$, the following properties hold:

(i) (Non-negativity) $d(x, y) \geq 0$

(ii) (Identity) $d(x, y) = 0 \iff x = y$

(iii) (Symmetry) $d(x, y) = d(y, x)$

(iv) (Triangle inequality) $d(x, z) \leq d(x, y) + d(y, z)$

We list down some metric spaces below.

Example 1.1.1.1.

(i) **Real line** with standard distance: $(\mathbb{R},\ d(x, y) = |x - y|)$; we will denote this metric space usually as $(\mathbb{R}, |\cdot|)$.

(ii) **Euclidean metric** (2-norm): $(\mathbb{R}^n,\ d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2})$

(iii) **Euclidean metric** (2-norm) on $\mathbb{C}^n$: $(\mathbb{C}^n,\ d(z, w) = \sqrt{\sum_{i=1}^n |z_i - w_i|^2})$

(iv) **Manhattan (taxicab) metric** (1-norm): $(\mathbb{R}^n,\ d(x, y) = \sum_{i=1}^n |x_i - y_i|)$

(v) **Maximum (supremum) metric** (∞-norm) : $(\mathbb{R}^n,\ d(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|)$:

(vi) **Discrete metric space**: On any set X $(X,\ d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases})$:

(vii) **Graph metric:** (Vertex set of a graph, $d(x, y) =$ length of shortest path)

(viii) **Geodesic arc-lengt**h on unit circle S^1: $(S^1,\ d(\theta_1, \theta_2) = \min(|\theta_1 - \theta_2|, 2\pi - |\theta_1 - \theta_2|))$:

(ix) **Spherical metric** on $S^1 \subset \mathbb{R}^2$: Let $\|\cdot\|_2$ denote the euclidean metric induced by the euclidean inner product $\langle \cdot, \cdot \rangle$ Then: $(S^1,\ d(x, y) = \arccos \dfrac{\langle x, y \rangle}{\|x\|\|y\|}$ Note the range of arccos is fixed to be $[0, \pi]$.

(x) Logarithmic metric on positive real numbers $(\mathbb{R}^+,\ d(x, y) = |\log(x/y)|)$:

(xi) $(S^{n-1} \subset \mathbb{R}^n,\ d(x, y) = \sqrt{2(1 - \langle x, y \rangle)})$: **Angular (cosine) distance on unit sphere**

(xii) **Binary Strings (Hamming Distance)**: Set $X = \{0,1\}^n$, the set of binary strings of length n. Metric: $d(x,y) = \sum_{i=1}^{n} |x_i - y_i|$

(xiii) **Infinite binary Strings**: $X = \{0,1\}^{\mathbb{N}} := \{f : \mathbb{N} \to \{0,1\}\}$ the collection of all sequences of $0,1$. Metric on X as

$$d_H(f,g) = \sum_{n=1}^{\infty} \frac{1}{2^n} |f(n) - g(n)|$$

Another metric on this space is

$$d_C(f,g) := 2^{-N(f,g)} \text{ where } N(f,g) = \min_{n \in \mathbb{N}} |f(n) - g(n)| \neq 0 \text{ for } f \neq g;$$

and

$$d_C(f,f) := 0, \quad f \in X = \{0,1\}^{\mathbb{N}}.$$

(xiv) **Feature Vectors** (Mahalanobis Distance): Set $X \subset M_{n\times 1}(\mathbb{R})$, fix some $S \in M_n(\mathbb{R})$ such that $\det(S) \neq 0$. Metric

$$d(X,Y) = \sqrt{(X-Y)^{\top} S^{-1} (X-Y)}$$

Example 1.1.1.2. Let (X,d) be a metric space. And let $\emptyset \neq M \subseteq X$. Then the restriction of the metric function d on M is a metic on M.

Definition 1.1.2 (Subspace of Metric space)**.** Let (X,d) be a metric space and leg $M \subseteq X$ be a non-empty subset. Then (M,d) is called a metric subspace of (X,d).

Example 1.1.2.1. The metric spaces $(\mathbb{Q}, |\cdot|)$, $(\mathbb{N}, |\cdot|)$ and $(\mathbb{R}, |\cdot|)$ are subspaces of $(\mathbb{C}, |\dot{}|)$.

Problem 1.1.3. Let (X,d) be a metric space. Then show that

$$|d(x,z) - d(y,z)| \leq d(x,y), \quad x,y,z \in X.$$

Hint: use triangle inequality.

Problem 1.1.4. Let d_1, d_2 be to metric on a set X. Then show that $d_1 + d_2$ is also metric on X.

Problem 1.1.5. Give a example to show that $d_1 \times d_2$ need not be a metric.

Definition 1.1.6 (Diameter of a metric space)**.** Let (X,d) be a metric space. Let $A \subseteq X$ then the diameter of A in (X,d) is

$$\operatorname{diam}_d(X) = \sup_{x,y \in A} d(x,y).$$

with the convention that $\operatorname{diam}(\varnothing) = 0$.

Problem 1.1.7. Let $A \subseteq X$ then show that

$$\operatorname{diam}_d(A) \leq \operatorname{diam}_d(X).$$

Problem 1.1.8. We verify the following metric on $\mathbb{R}$

$$d(x, y) = \frac{|x - y|}{1 + |x - y|}$$

With The metric d not $\operatorname{diam}_d(\mathbb{R}) = 1$. We will call d_b the bounded metric on $\mathbb{R}$ induced by the metric d.

Definition 1.1.9 (Bounded Metric Space)**.** Let (X, d) be a metric space. Then it is called a bounded metric space if

$$\operatorname{diam}_d(X) < \infty,$$

that is $\operatorname{diam}_d(X)$ is finite.

Definition 1.1.10 (Distance between sets)**.** Let (X, d) be a metric space and $A, B \subset X$. Then the distance between A, B is defined to be

$$\operatorname{dist}(A, B) = \inf\{d(x, y) \mid x \in A, y \in B\}$$

—

Let (X, d) be a bounded metric space. For any subsets $A, B \subseteq X$, define

$$\rho(A, B) = \operatorname{diam}(A \triangle B),$$

where $A \triangle B = (A \setminus B) \cup (B \setminus A)$ is the *symmetric difference* of A and B.

Note the following properties hold for diam.

(i) $\rho(A, B) \geq 0, \quad \text{and} \quad \rho(A, A) = 0.$

(ii) $\rho(A, B) = \operatorname{diam}(A \triangle B) = \operatorname{diam}(B \triangle A) = \rho(B, A).$

(iii) For every $A, B, C \subseteq X$,

$$\begin{aligned} \rho(A, C) &= \operatorname{diam}(A \triangle C) \\ &\leq \operatorname{diam}\big((A \triangle B) \cup (B \triangle C)\big) \\ &\leq \max\{\operatorname{diam}(A \triangle B), \operatorname{diam}(B \triangle C)\} \\ &= \max\{\rho(A, B), \rho(B, C)\}. \end{aligned}$$

In particular, ρ satisfies an identity that is stronger than triangle inequality.

But, in general, ρ is not a true metric, since

$$\rho(A,B)=0 \;\;\Rightarrow\;\; \operatorname{diam}_d(A\triangle B)=0.$$

But, $\operatorname{diam}_d(A\triangle B)=0$ need not imply that $A\triangle B=\emptyset$.

Let $(X,d)=(\mathbb{R},|\cdot|)$ be the real line with the usual metric and define $\rho(A,B)=\operatorname{diam}(A\triangle B)$ for $A,B\subset\mathbb{R}$. Take

$$A=\{0\}, \qquad B=\{0,1\}.$$

Then the symmetric difference is the singleton

$$A\triangle B=\{1\},$$

and therefore

$$\rho(A,B)=\operatorname{diam}_{|\cdot|}(\{1\})=\sup\{\,|x-y| : x,y\in\{1\}\,\}=0.$$

But $A\neq B$. Hence $\rho(A,B)=0$ while A and B are distinct — so ρ is not a metric on $\mathcal{P}(\mathbb{R})$.

Definition 1.1.11 (Pseudometric)**.** Let X be a nonempty set. A function

$$d : X\times X\to[0,\infty)$$

is called a *pseudometric* on X if it satisfies the following properties for all $x,y,z\in X$:

1. $d(x,y)\geq 0$.
2. $d(x,x)=0$. Note that we do *not* require $d(x,y)=0\Rightarrow x=y$.
3. $d(x,y)=d(y,x)$.
4. $d(x,z)\leq d(x,y)+d(y,z)$.

If, in addition, the implication

$$d(x,y)=0 \;\Rightarrow\; x=y$$

holds, then d is becomes a *metric*.

Definition 1.1.12 (Ultrametric)**.** Let X be a nonempty set. A function

$$d : X\times X\to[0,\infty)$$

is called an *ultrametric* on X if it satisfies, for all $x,y,z\in X$,

1. $d(x,y)\geq 0$.
2. $d(x,y)=0 \iff x=y$.

3. $d(x, y) = d(y, x)$.

4. For any $x, y, z \in X$ the following equality hold:

$$d(x, z) \leq \max\{\, d(x, y),\ d(y, z)\,\}.$$

So we get a pseudometric by relaxing the identity condition and we get a ultrametric by strengthening the triangle inequality. So $\rho_d(A, B) = \mathrm{diam_d}(A \triangle B)$ is an ultra-pseudo-metric on $\mathcal{P}(X)$.

Problem 1.1.13. Show that on $X = \{0, 1\}^{\mathbb{N}} := \{f \ : \ \mathbb{N} \to \{0, 1\}\}$ the metric

$$d_C(f, g) := 2^{-N(f,g)} \text{ where } N(f, g) = \min_{n \in \mathbb{N}} |f(n) - g(n)| \neq 0 \text{ for } f \neq g;$$

and

$$d_C(f, f) := 0, \quad f \in X = \{0, 1\}^{\mathbb{N}}.$$

is an ultra-metric.

The following figure demonstrates how the sets

$$T_p = \left\{(x_1, x_2) \in \mathbb{R}^2 \;\mid\; |x_1|^p + |x_2|^p = 1\right\} \text{ and } T_\infty = \left\{x \in \mathbb{R}^2 \mid \max\{x_1, x_2\} = 1\right\}.$$

are visualized on the coordinate plane for $p = 1, 2, 3, 4, 1/2, 1/3, 1/4$. Note the difference between $p \geq 1$ and $p < 1$.

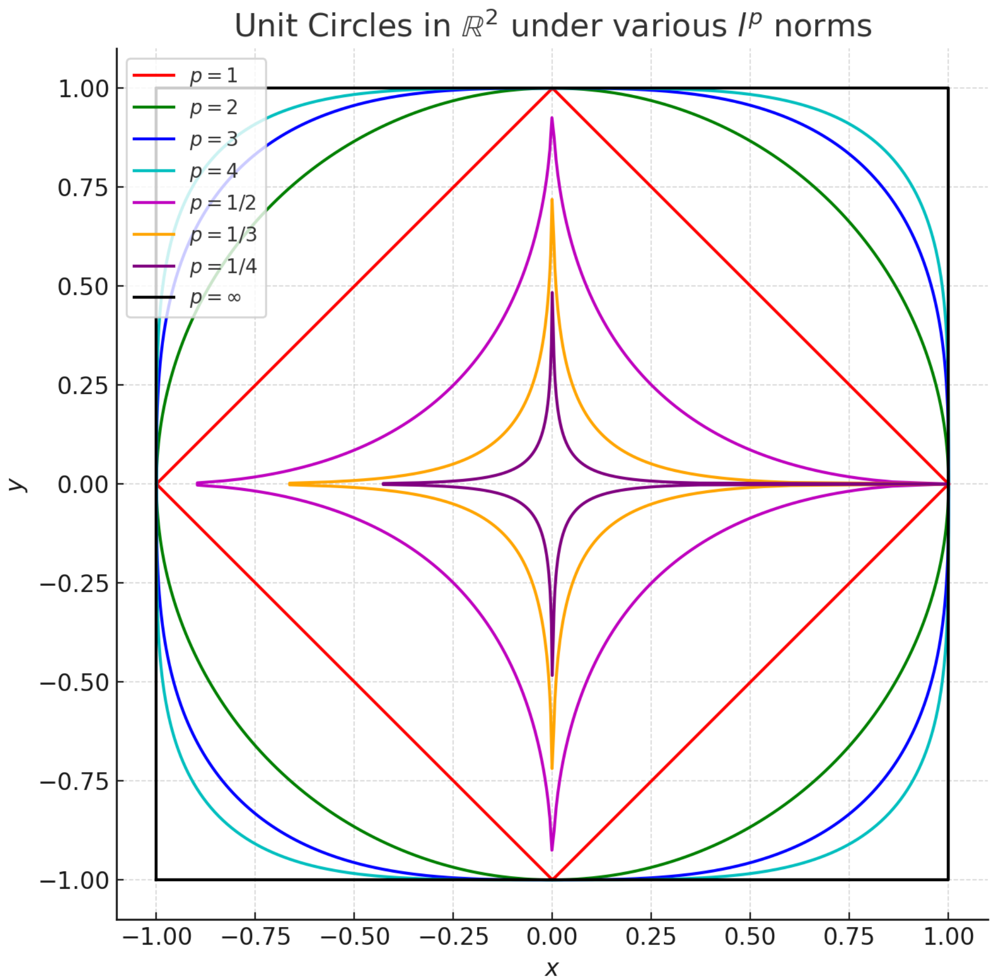

Figure 1.1: The set T_p in $\mathbb{R}^2$ under various p, including $p = 1, 2, 3, 4, \infty$ and fractional $p = \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$.

Example 1.1.13.1. We show with an example that triangle inequality fails for $\mathbb{R}^2$ for

$$d_p(x, y) = \left(|y_1 - x_1|^p + |y_2 - x_2|^p\right)^{1/p}, \quad 0 < p < 1.$$

Consider the three points in $\mathbb{R}^2$

$$x = (0, 0), \qquad y = (1, 0), \qquad z = (1, 1).$$

Then

$$d_p(x, y) = (|1-0|^p + |0-0|^p)^{1/p} = (1^p + 0)^{1/p} = 1,$$

$$d_p(y, z) = (|1-1|^p + |1-0|^p)^{1/p} = (0 + 1^p)^{1/p} = 1,$$

$$d_p(x, z) = (|1-0|^p + |1-0|^p)^{1/p} = (1^p + 1^p)^{1/p} = (2)^{1/p}.$$

Since $0 < p < 1$, we have $1/p > 1$, hence $2^{1/p} > 2$. Therefore,

$$d_p(x, z) = 2^{1/p} > 2 = d_p(x, y) + d_p(y, z),$$

which shows that the *triangle inequality fails* for d_p when $p < 1$. So for, $p < 1$, d_p is not a metric on $\mathbb{R}^2$.

1.2 Metric space Topology

Definition 1.2.1 (Open Ball)**.** Let (X, d) be a metric space. For any $x \in X$ and $\varepsilon > 0$, the **open ball** centered at x with radius ε is defined by:

$$B_d(x, \varepsilon) = \{y \in X : d(x, y) < \varepsilon\}.$$

Definition 1.2.2 (Open Set and Closed Set)**.** A subset $U \subseteq X$ is called **open** if for every $x \in U$, there exists $\varepsilon > 0$ such that:

$$B_d(x, \varepsilon) \subseteq U.$$

That is, U is open if it contains an open ball around each of its points. A set is $E \subseteq X$ is said to be closed if E^c is open in X.

Example 1.2.2.1. Let (X, d) be a metric space. Then the sets $X, \emptyset$ are trivially both open and closed sets.

Example 1.2.2.2. Every open interval in $\mathbb{R}$ is a open set in $(\mathbb{R}, |\cdot|)$.

Example 1.2.2.3. Let (X, d) be the discrete metric space then every $\{x\} \subset X$ is an open set.

Definition 1.2.3 (Isolated Point)**.** Let $A \subseteq X$. A point $x \in A$ is called an **isolated point** of A if there exists $\varepsilon > 0$ such that:

$$B_d(x, \varepsilon) \cap A = \{x\}.$$

That is, x is isolated in A if it does not have any other points of A arbitrarily close to it.

Example 1.2.3.1. Show that in discrete metric space (X, d) defined in every point in X is isolated.

Theorem 1.2.4 (Arbitrary union of open sets)**.** Let (X, d) be a metric space. Then, for any collection $\{U_i\}_{i\in I}$ of open sets the union $\cup_{i\in I} U_i$ is an open set in X.

Proof. If each $U_i = \emptyset$. Then there is nothing to prove. So without loss of generality assume that at least one of $U_i \neq \emptyset$.

Let $x \in \cup i \in IU_i$, then there exist some $i_0 \in I$ such that $x \in U_{i_0}$. Since U_{i_0} is open, there exist a $\varepsilon > 0$ such that

$$B_\varepsilon(x_0) \subset U_{i_0} \subset \bigcup_{i\in I} U_i.$$

Hence $\cup_{i\in I} U_i$ is an open set in X. ■

Corollary 1.2.4.1. Let (X, d) be a metric space. Then, for any collection $\{E_i\}_{i\in I}$ of closed sets the intersection $\cup_{i\in I} E_i$ is a closed set in X.

Proof. By definition E_i is closed hence E_i^c is open in X. Hence by the above theorem

$$\bigcup_{i\in I} E_i^c \text{ is an open set.}$$

Hence

$$\left(\bigcup_{i\in I} E_i\right)^c \text{ is an open set.}$$

By definition of closed set

$$\bigcup_{i\in I} E_i \text{ is a closed set.}$$

■

Corollary 1.2.4.2. Let (X, d) be a metric space. Then, for any collection $\{E_i \mid i \in I\}$ of closed sets the intersection $\cap_{i\in I} E_i$ is a closed set in X.

Theorem 1.2.5 (Finite intersection)**.** Let (X, d) be a metric space. Then, for any finite collection $\{U_i \mid i = 1, \dots n\}$ of open sets then the intersection $\cap_{i=1}^n U_i$ is an open set in X.

Proof. If $\cap_{i=1}^n U_i = \emptyset$ then there is nothing to prove. WLOG assume that every $\cap_{i=1}^n U_i \neq \emptyset$.

Let $x \in \cap_{i=1}^n U_i$. Since each U_i is open there exits $\varepsilon_i > 0$ such that

$$B_{\varepsilon_i}(x) \subset U_i, \quad i = 1, \dots, n.$$

Let $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\}$. Hence

$$B_\varepsilon(x) \subset B_{\varepsilon_i}(x) \subset U_i, \quad i = 1, \dots, n.$$

So

$$B_\varepsilon(x) \subset \bigcap_{i=1}^n U_i.$$

Hence $\cap_{i=1}^n U_i$ is an open set. ■

Corollary 1.2.5.1 (Finite union of closed sets)**.** Let $E_1, \ldots, E_n$ be a (finite) collection of closed sets in a metric space (X, d). Then

$$\bigcap_{i=1}^{n} E_i \text{ is an closed set.}$$

Proof. Left as an exercise. ■

Theorem 1.2.6 (Hausdorff Point separation)**.** Let (X, d) be a metric space. Let $x_1, x_2 \in X$ be two distinct points (i.e.$x_1 \neq x_2$). Then there exists $\varepsilon_1 > 0, \varepsilon_2 > 0$ such that

$$B_{\varepsilon_1}(x_1) \bigcap B_{\varepsilon_2}(x_2) = \emptyset.$$

Proof. Let (X, d) be a metric space. Let $x_1, x_2 \in X$ be two distinct points (i.e.$x_1 \neq x_2$). Then

$$d(x_1, x_2) > 0.$$

Let $\varepsilon_1 = \dfrac{d(x_1, x_2)}{3}$ and $\varepsilon_1 = \dfrac{d(x_1, x_2)}{3}$ then we claim that

$$B_{\varepsilon_1}(x_1) \bigcap B_{\varepsilon_2}(x_2) = \emptyset.$$

If not then suppose $x \in B_{\varepsilon_1}(x_1) \bigcap B_{\varepsilon_2}(x_2)$. Then by triangle inequality

$$d(x_1, x_2) \leq d(x_1, x) + d(x_1, x) = 2\frac{d(x_1, x_2)}{3} < d(x_1, x_2)$$

So arrived at contradiction. Hence

$$B_{\varepsilon_1}(x_1) \bigcap B_{\varepsilon_2}(x_2) = \emptyset.$$

■

Problem 1.2.7. Let (X, d) be a metric space. Then the set $\{x\} \subseteq X$ is a closed set in X.

Problem 1.2.8. Show that $\mathbb{Q}$ is not closed in $(\mathbb{R}, |\cdot|)$.

Remark 1.2.9. Since Q is countable, it is countable union of single tons. In any metric space every single not set is closed. In conclusion, countable union of closed sets need not be closed.

Problem 1.2.10. Let (X, d) be a metric space and let (M, d) be a subspace of (X, d). Let $U \subseteq M$. Then show that

$$U \text{ is open in } M \Leftrightarrow \exists \quad O \subseteq X \text{ open in } (X, d) \text{ such that } U = M \cap O.$$

Problem 1.2.11. In a discrete metric space show that every set os simultaneously open and closed.

Example 1.2.11.1. Show that in the subspace metric on $X = (0,1) \cup \{-1\}$ of the metric space $(\mathbb{R}, |\cdot|)$ the set $(0,1)$ is both open and closed. Equivalently $\{-1\}$ is both open and closed.

Problem 1.2.12. Show that in the discrete metric space (X,d) every set is simultaneously open and closed.

Definition 1.2.13 (Convergent Sequence)**.** Let (X,d) be a metric space. A sequence $\{x_n\}_{n\in\mathbb{N}}$ is said to converge to $x \in X$ if $\forall \varepsilon > 0,\ \exists N \in \mathbb{N}$ such that

$$d(x_n, x) < \varepsilon, \quad n \geq N.$$

We shorthand the above as $\lim\limits_{n\to\infty} x_n = x$ or $x_n \to x$ in X.

Example 1.2.13.1. Find the limit of the sequence given by $x_n = 2^{-n}$. Find limit of $\{x_n\}_{n\in\mathbb{N}}$ in the metric space $(\mathbb{R}, d)$ where

$$d(x,y) = \frac{|x-y|}{1+|x-y|}.$$

Definition 1.2.14. A sequence $\{x_n\}$ in a metric space (X,d) is called Cauchy if for any $\varepsilon > 0$ there exists an integer $N > 0$ such that

$$d(x_n, x_m) < \varepsilon \text{ for all } , j \geqq N$$

.

Remark 1.2.15. In the notation of open balls, we can write a sequence $\{x_n\}_{n\in N}$ in X converges to $x \in X$ if for every $\varepsilon > 0$ there is a $N \in \mathbb{N}$ such that

$$x_n \in B_d(x, \varepsilon) \quad n \geq N.$$

Notation: If (X,d) is a metric space and $\{x_n\}_{n\in\mathbb{N}}$ is a sequence converging to $x \in X$ then we shorthand this as

$$x_n \xrightarrow{d} x$$

In case X is clear from the context we will use the above notion without mentioning X.

Problem 1.2.16. Show that $x_n \xrightarrow{d} x$ if and only if $d(x_n, x) \xrightarrow{|\cdot|} 0$.

Definition 1.2.17 (Limit points)**.** Let (X,d) be a metric space. A set $E \subset X$. A point $x \in X$ is said to be a limit point of E if

$$B_\varepsilon(x) \setminus \{x\} \bigcap E \neq \emptyset, \quad \varepsilon > 0.$$

Problem 1.2.18. Let (X,d) be a metric space. A set $E \subset X$. Show that $x \in X$ is a limit point of E then there exists a sequence $\{x_n\}_{n\in N} \subset E$ o such that

$$\lim x_n = x.$$

Theorem 1.2.19. Let (X,d) be a metric space. A set $E \subset X$ is closed in X if and only if limit point of E is contained in E.

Proof. Let $E \subset X$ is closed in X.
If the collection of limit points of E is not $\emptyset$. Then the the statement vacuously holds. Now suppose x be a limit point of E. We claim that $x \in E$.

If not, then $x \in E^c$. By definition of E^c is an open set. Hence there exist $\varepsilon > 0$ such that $B_\varepsilon(x) \subset E^c$. Thus

$$B_\varepsilon(x) \cap E = \emptyset.$$

hence

$$B_\varepsilon(x) \setminus \{x\} \cap E = \emptyset.$$

This contradicts the fact that x is a limit point. Hence

$$\lim_n x_n = x.$$

Hence the claim stands and $x \in E$.

Conversely assume that E contains all its limit point. Case-1: The set of limit points of E is non-empty.

We claim that E is closed. Let $x \in E^c$.
Then we claim that there exist $\varepsilon > 0$ such that

$$B_\varepsilon(x) \subset E^c.$$

If not then for every $\varepsilon > 0$

$$B_\varepsilon(x) \cap E \neq \emptyset.$$

But $x \in E^c$ so by the above inclusion implies

$$B_\varepsilon(x) \setminus \{x\} \cap E \neq \emptyset.$$

Hence x is a limit point of E. So by hypothesis $x \in E$. This contradicts our assumption that $x\ inE^c$.

Hence there exist $\varepsilon > 0$ such that

$$B_\varepsilon(x) \subset E^c.$$

Case-2: The set of limit points of E is empty.
Claim that E is closed equivalently E^c open. If not then $x \in E^c$ such that

$$\forall \varepsilon > 0 \quad B_\varepsilon(x) \cap E \neq \emptyset.$$

This implies x is limit point of E. This contradicts the fact that the set of limit points of E is empty. Hence E^c is open equivalently E is closed. This completes the proof. ∎

Remark 1.2.20. Warning: Limit points of a set and limit of a sequence should not be confused with. For example $\{x_n = (-1)^n + \frac{1}{n}\}_{n\in\mathbb{N}} \subset \mathbb{R}$. Then with respect to the usual metric of $\mathbb{R}$ there is no limit to the sequence $\{x_n\}$. But the set of limit points is $\{-1, 1\}$.

Definition 1.2.21 (Cauchy Sequence)**.** Let (X, d) be a metric space. A sequence $x_{n n\in\mathbb{N}}$ in X is said to be a Cauchy sequence if

$$\forall\, \varepsilon > 0,\ \exists\, N \in \mathbb{N} \text{ such that } \forall\, m, n \geq N,\ d(x_m, x_n) < \varepsilon.$$

Theorem 1.2.22. Any convergent sequence in a metric space (X, d) is a Cauchy sequence.

Proof. Suppose $x_n \xrightarrow{d} x$. Given ε, pick N large enough so that $d\left(x_n, x\right) < \varepsilon/2$ for $n \geqq N$. Then for $n, m \geqq N$,

$$d\left(x_n, x_m\right) \leqq d\left(x_n, x\right) + D\left(x, x_m\right) < \varepsilon/2 + \varepsilon/2 = \varepsilon$$

∎

Definition 1.2.23 (Sub sequence)**.** Let $\{x_n\}_{n\in\mathbb{N}}$ be sequence in (X, d). If $n_1 < n_2 < \ldots < n_k < \ldots <$ be a strictly increasing sequence of natural numbers then

$$\{x_{n_k}\}_{k\in\mathbb{N}}$$

is called a subsequence of $\{x_n\}$.

Problem 1.2.24. Let x be limit point of the set $\{x_n \ : \ n \in \mathbb{N}\}$ then show that there exists a subsequence $\{x_{n_k}\}_{k\in\mathbb{N}}$ of the sequence $\{x_n\}_{n\in\mathbb{N}}$ that converges to $x_{n_k} \to x$.

Problem 1.2.25. Let $\{x_n\}_{n\in\mathbb{N}}$ be a Cauchy sequence in a metric space (X, d). Let $\{x_{n_k}\}_{k\in\mathbb{N}}$ be a subsequence of $\{x_n\}$. If $x_{n_k} \xrightarrow{d} x$, then show that $x_n \xrightarrow{d} x$.

Example 1.2.25.1. Let $\{x_n\}_{n\in\mathbb{N}}$ be a Cauchy sequence in (X, d). Then show that the diameter of the set $\{x_i \ : \ i \in \mathbb{N}\}$ is finite .

Example 1.2.25.2. Let $\{x_n\}_{n\in\mathbb{N}}$ be a Cauchy sequence such that the set $\{x_n \ : \ n \in \mathbb{N}\}$ is a finite set then show that $\{x_n\}_{n\in\mathbb{N}}$ is convergent.

Let $\{x_n\}_{n\in\mathbb{N}}$ be a Cauchy sequence and suppose the range

$$S := \{x_n : n \in \mathbb{N}\}$$

is finite. If S has only one element then $\{x_n\}_{n\in\mathbb{N}}$ is constant and hence convergent. So assume S contains at least two distinct points, say

$$S = \{a_1, \ldots, a_k\}, \qquad k \geq 2,$$

where the a_i are distinct. Because the set of pairwise distances between distinct a_i is finite and all positive, the quantity

$$\delta := \min_{1\leq i<j\leq k} d(a_i, a_j)$$

is well-defined and $\delta > 0$.

Since $\{x_n\}_{n\in\mathbb{N}}$ is Cauchy, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$,

$$d(x_m, x_n) < \frac{\delta}{2}.$$

If for some $m, n \geq N$ we had $x_m \neq x_n$, then $x_m = a_i$ and $x_n = a_j$ for some $i \neq j$, whence

$$d(x_m, x_n) = d(a_i, a_j) \geq \delta,$$

a contradiction with $d(x_m, x_n) < \delta/2$. Therefore all terms x_n with $n \geq N$ are equal to the same element $a \in S$; i.e. the sequence is eventually constant. An eventually constant sequence converges (to that constant), so $\{x_n\}_{n\in\mathbb{N}}$ is convergent.

Definition 1.2.26 (Complete metric space)**.** A metric space M is complete if every Cauchy sequence in M is convergent in M.

Theorem 1.2.27. Let (X, d) be a complete metric space. A subset $A \subseteq X$ is closed in X if and only if (A, d) is a complete metric space.

Proof. Let $\{x_n\}$ be a Cauchy sequence in A. It can be regarded as a Cauchy sequence in X, and as such it converges to an element x in X. Since A is closed $x \in A$. Hence A is complete.

Conversely, let A be complete. By ∎

Theorem 1.2.28. Let (X, d) be a complete metric space. If $A \subseteq X$ be such that (A, d) is complete if and only if A is a closed subset of X.

Proof. Given $\{x_i\}$ in A converging to $x \in X$, we have to show that $x \in A$. Now the sequence $\{x_i\}$ is Cauchy. Since A is complete, $\{x_i\}$ converges to a limit in A. Since limit of any sequence is unique in a metric space, $x \in A$. ∎

Problem 1.2.29. Let (X, d) be metric space and let $\{A_n\}$ be a descending sequence of closed subsets of X; ie $A_1 \supset A_2 \supset A_3 \supset \cdots$. If diam $(A_i) \to 0$, then show that $\cap A_i$ is either empty or contains exactly one point. Gives examples showing either of the scenario may occur.

Example 1.2.29.1. $(\mathbb{R}, |\cdot|)$ is a complete metric space. But neither $(\mathbb{Q}, |\cdot|)$ nor $(\mathbb{Z}, |\cdot|)$ is complete.

Example 1.2.29.2. Let (X, d) be a discrete metric space, then (X, d) is complete. Note, if $\{x_n\}_{n\in\mathbb{N}}$ is Cauchy sequence in X then for $\varepsilon = 1/2$ there exist $N \in \mathbb{N}$ such that

$$d(x_m, x_n) < \frac{1}{2}, \quad m, n > N.$$

But $d(x_m, x_n) < 1/2$ implies $d(x_m, x_n) = 0$. So, we have shown that there exists $N \in \mathbb{N}$ such that

$$x_n = x_m, \quad n, m > N.$$

Such a sequence is called an eventually constant sequence. Clearly an eventually constant sequence is convergent.

Definition 1.2.30 (Closure)**.** Let (X, d) be a metric space. Then the closure of A, denoted by $\overline{A}$, is the smallest closed set containing A. That is, if $A \subseteq S$ and S is closed then $\overline{A} \subset S$.

Problem 1.2.31. Show that $A \subset X$ is closed in (X, d) if and only if $A = \overline{A}$.

Theorem 1.2.32. Let (X, d) be a metric space and let $A \subset X$. Then show that the following are equivalent

(i) $x \in \overline{A}$.

(ii) For every $\varepsilon > 0$

$$B_\varepsilon(x) \cap A \neq \emptyset.$$

(iii) There exists a sequence $\{x_n\}_{n \in \mathbb{N}} \subset A$ such that

$$\lim_{n \to \infty} x_n = x.$$

Proof. We only show that $(i) \Rightarrow (ii)$ rest is left as exercise. Let $x \in \overline{A}$. We claim that For every $\varepsilon > 0$

$$B_\varepsilon(x) \cap A \neq \emptyset.$$

Case:i If $x \in A$ then there is noting to prove. WLOG we can assume that $A \neq \overline{A}$. Assume that $x \in \overline{A} \setminus A$. If the claim fails, then there exists $\varepsilon > 0$ such that

$$B_\varepsilon(x) \cap A \neq \emptyset \Rightarrow B_\varepsilon(x) \subset A^c.$$

But

$$B_\varepsilon(x) \subset A^c \Longleftrightarrow A \subset B_\varepsilon(x)^c.$$

But $B_\varepsilon(X)^c$ is a closed set. Hence the above inclusion implies

$$\overline{A} \subset B_\varepsilon(x)^c.$$

This gives contradiction as

$$x \in \overline{A} \subset B_\varepsilon(x)^c;$$

but obviously $x \notin B_\varepsilon(x)^c$. Hence (ii) must hold.

Rest of the equivalences are left as an exercise. ∎

Corollary 1.2.32.1. Let $A \subset X$ then

$$\overline{A} \setminus A = \{x \in A^c \ : \ \forall \varepsilon > 0, \quad (B_\varepsilon(x) \setminus \{x\}) \cap A \neq \emptyset\}$$

That is $\overline{A} \setminus A$ is the collection of limit points of A that are not in A.

Problem 1.2.33. Let (X, d) be metric space. Let $A \subseteq X$, then show that

$$\operatorname{diam}(A) = \operatorname{diam}(\overline{A}).$$

Proof. Recall that for any subset $B \subseteq X$, the diameter of B is defined by

$$\operatorname{diam}(B) = \sup\{d(x, y) : x, y \in B\}.$$

Since $A \subseteq \overline{A}$, we have

$$\operatorname{diam}(A) \leq \operatorname{diam}(\overline{A}).$$

For the reverse inequality, let $x, y \in \overline{A}$. If $\operatorname{diam}(A) = \infty$, then clearly $\operatorname{diam}(\overline{A}) = \infty$, so assume $\operatorname{diam}(A) < \infty$.

Fix $\varepsilon > 0$. Since $x, y \in \overline{A}$, there exist points $a, b \in A$ such that

$$d(x, a) < \frac{\varepsilon}{3} \quad \text{and} \quad d(y, b) < \frac{\varepsilon}{3}.$$

Then by the triangle inequality,

$$\begin{aligned} d(x, y) &\leq d(x, a) + d(a, b) + d(b, y) \\ &< \frac{\varepsilon}{3} + d(a, b) + \frac{\varepsilon}{3} = d(a, b) + \frac{2\varepsilon}{3} \\ &\leq \operatorname{diam}(A) + \frac{2\varepsilon}{3}. \end{aligned}$$

Since this holds for all $\varepsilon > 0$, we get $d(x, y) \leq \operatorname{diam}(A)$.

Taking the supremum over all $x, y \in \overline{A}$ gives

$$\operatorname{diam}(\overline{A}) \leq \operatorname{diam}(A).$$

Hence,

$$\operatorname{diam}(A) = \operatorname{diam}(\overline{A}).$$

■

Definition 1.2.34 (Isolated point)**.** Let (X, d) be a metric space and let $A \subset X$. A point in A is called and isolated point if it is not a limit point of A.

Example 1.2.34.1. In the metric space $(\mathbb{R}, |\cdot|)$ there are $\mathbb{Q}$ does not have any isolated points. I The set $A = \{\frac{1}{n} \ : \ n \in \mathbb{N}, n \geq 1\} \cup \{0\}$ is a closed set. The set $\{\frac{1}{n} \ : \ n \in \mathbb{N}, n \geq 1\}$ is set of isolated points in A, and $\{0\}$ is only limit point.

Problem 1.2.35. Let (X, d) be a metric space . Let $x \in X$ be an isolated point of X. Then show that $\{x\}$ both open and closed.

Example 1.2.35.1. In the subspace metric $(A, |\cdot|)$ of $(\mathbb{R}, |\cdot|)$ where $A = \{\frac{1}{n} \ : \ n \in \mathbb{N}, n \geq 1\} \cup \{0\}$ every proper subset of $A = \{\frac{1}{n} \ : \ n \in \mathbb{N}, n \geq 1\} \cup \{0\}$, is both open and closed.

Definition 1.2.36 (Boundary of a set)**.** Let (X, d) be a metric space then boundary of any $A \subseteq X$ is defined to be

$$\{x \in X \ : \ \forall \varepsilon > 0 \ \ B_\varepsilon(x) \cap A \neq \emptyset \text{ and } B_\varepsilon(x) \cap A^c \neq \emptyset\}.$$

We short hand the boundary of a set A as ∂A.

Proposition 1.2.37. Let (X, d) be a metric space. Then for any $A \subseteq X$,

$$\partial A = \overline{A} \cap \overline{A^c}.$$

Proof. Follows from the definition. ■

Problem 1.2.38. Let (X, d) be a metric space, the show that for any $A \subset X$ show that $\overline{A} \setminus A \subseteq \partial A$.

Example 1.2.38.1. In the metric space $(\mathbb{R}, |\cdot|)$, consider the set $A = [0, 1] \cup \{2\}$. Then

1. $\partial A = \{0, 1, 2\}$.
2. $\overline{A} = A$.
3. $\overline{A} \setminus A = \emptyset$.
4. Set of limit points of A is $[0, 1]$.
5. $\{2\}$ is an isolated point.

Problem 1.2.39. In $(\mathbb{R}, |\cdot|)$ show that of any $A \subseteq \mathbb{R}$, if $x \in A$ is an isolated point then $x \in \partial A$.

Example 1.2.39.1. Let (X, d) be the discrete metric space. Let $A \subseteq X$. Then

$$\overline{A} = A$$

and

$$\overline{A^c} = A^c.$$

Hence

$$\partial A = A \cap A^c = \emptyset.$$

Example 1.2.39.2. In the discrete metric space (X, d), if $A \subset X$ is a nonempty set then every $x \in A$ is an isolated point in A. But $\partial(A) = \emptyset$. This provides an example of metric space where isolated points need not be in the boundary of a set.

The set above is sometime denoted as

$$A' = \overline{A} \setminus A.$$

And A' is called the derived set of A. This is sometimes referred to as the Cantor-Bendixon derivative.

Definition 1.2.40 (Interior Points)**.** Let (X, d) be metric space and $A \subseteq X$. Then interior of A is the largest open set contained in A. We denote A° to be interior of A. In other words, $A^\circ \subseteq X$ is an open set such that,

$$U \subseteq A \text{ is any open set then } U \subseteq A^\circ \subseteq A.$$

Note, if we collect all open sets contained in A, say as A_O, then the collectionA_O, is non-empty as $\emptyset \in A_O$. Moreover,

$$\bigcup_{U \in A_O} U \subset A.$$

Since each $U \in A_O$ is open and arbitrary union of open sets are open, $\bigcup_{U \in A_O} U$ is open.

Problem 1.2.41. Show that $\bigcup_{U \in A_O} U = A^\circ$.

Problem 1.2.42. Show that $A \subset X$ is open in (X, d) if and only if $A = A^\circ$.

Proposition 1.2.43. Let (X, d) be a metric space. And let $A \subseteq X$, then show that

$$\{x \in X \;:\; \text{for } x \;\; \exists \;\; \varepsilon > 0 \quad B_{\varepsilon_x}(x) \subset A\} = A.$$

Proof. By definition $A^\circ \subseteq A$ is an open set . Hence for any $x \in A^\circ$ there exist ε such that

$$B_\varepsilon(x) \subseteq A^\circ \subset A.$$

Hence

$$A^\circ \subseteq \{x \in X \;:\; \text{for } x \;\; \exists \;\; \varepsilon > 0 \quad B_{\varepsilon_x}(x) \subset A\}. \tag{2.1}$$

Conversely, note

$$\{x \in X \;:\; \text{for } x \;\; \exists \;\; \varepsilon > 0 \quad B_{\varepsilon_x}(x) \subset A\} \subset A. \tag{2.2}$$

And we also claim that $\{x \in X \;:\; \text{for } x \;\; \exists \;\; \varepsilon > 0 \quad B_{\varepsilon_x}(x) \subset A\}$ is an open set.

Let $x \in \{x \in X \ : \ \text{ for } x \ \exists \ \ \varepsilon > 0 \ \ \ B_\varepsilon(x) \subset A\}$, then for any $y \in B_\varepsilon(x) \setminus \{x\}$ fix $r = \dfrac{d(x,y)}{3}$

$$B_r(y) \subseteq B_\varepsilon(x) \subseteq A.$$

Hence

$$B_\varepsilon(x) \subseteq \{x \in X \ : \ \text{ for } x \ \exists \ \ \varepsilon > 0 \ \ \ B_\varepsilon(x) \subset A\},$$

and thus

$$\{x \in X \ : \ \text{ for } x \ \exists \ \ \varepsilon > 0 \ \ \ B_\varepsilon(x) \subset A\} \text{ is an open set.} \tag{2.3}$$

By eq 2.2 and 2.3 we get

$$\{x \in X \ : \ \text{ for } x \ \exists \ \ \varepsilon > 0 \ \ \ B_\varepsilon(x) \subset A\} \subseteq A^\circ \subset A.$$

So we get

$$\{x \in X \ : \ \text{ for } x \ \exists \ \ \varepsilon > 0 \ \ \ B_\varepsilon(x) \subset A\} = A^\circ.$$

■

Proposition 1.2.44. Let (X,d) be a metric space and $A \subseteq X$ then we have the following identities hold

1. $A^\circ \cap \partial A = \emptyset$.
2. $\overline{A} = A^\circ \sqcup \partial A$.
3. $X = A^\circ \sqcup \partial A \sqcup (A^c)^\circ$.
4. $X = A^\circ \sqcup \partial(A^\circ) \sqcup (A^c)^\circ$.
5. $\partial(A) = \partial(A^c)$.

Proof. We first claim that $A^\circ \cap \partial A = \emptyset$. If suppose that the claim is not true, then let $x \in A^\circ \cap \partial A$.

Since $x \in A^\circ$ by above proposition there exists a $r > 0$ such that

$$B_r(x) \subseteq A.$$

Using $x \in \partial A$ we get

$$B_\varepsilon(x) \cap A^\circ \neq \emptyset.$$

This is gives a contradiction. Hence

$$A^\circ \cap \partial A = \emptyset.$$

Since

$$A^\circ \subseteq A \subseteq \overline{A}, \text{ and } \partial A = \overline{A} \cap \overline{A^c},$$

we get

$$A^\circ \sqcup \partial A \subseteq \overline{A}.$$

Conversely, let $x \in \overline{A} \setminus A^\circ$. By Theorem 1.2.32 for every $\varepsilon > 0$

$$B_\varepsilon(x) \cap A \neq \emptyset \quad \text{and} \quad B_\varepsilon(x) \not\subset A.$$

Hence

$$B_\varepsilon(x) \cap A^\circ \neq \emptyset.$$

So $x \in \partial A$. Hence

$$A^\circ \sqcup \partial A = \overline{A}.$$

Clearly, by symmetry of above equation

$$X = A^\circ \sqcup \partial(A^\circ) \sqcup (A^c)^\circ.$$

In particular,

$$A^\circ \sqcup \partial(A^\circ) \sqcup (A^c)^\circ =$$

■

Example 1.2.44.1. $\mathbb{Q}^\circ = \emptyset$ in the metric space $(\mathbb{R}, |\cdot|)$. Note , if $x \in \mathbb{R}$ for any $\varepsilon > 0$,

$$(x - \varepsilon, x + \varepsilon) \cap \mathbb{Q}^c \neq \emptyset.$$

So, $\mathbb{Q}^\circ = \emptyset$.

Example 1.2.44.2. Show that if A is an open set in some metric space (X, d) then $\partial A = \emptyset$.

Example 1.2.44.3. Let (X, d) be a metric space and let $A \subseteq X$ be any set. Then show that ∂A is a closed set in X.

Example 1.2.44.4. Consider the set $X = (0, 1) \cup \{2\}$ in the subspace metric of $(\mathbb{R}, |\cdot|)$. In the metric space $(X, |\cdot|)$. Then,

(i) $(0, 1)$ and $\{2\}$ bot sets are closed as well as open in $(X, |\cdot|)$.

(ii) $\partial(0, 1) = \emptyset$ and $\partial(\{2\}) = \emptyset$.

(iii) $\partial(\mathbb{Q} \cap (0, 1)) = (0, 1)$

Definition 1.2.45 (Dense Set)**.** Let (X, d) be a metric space. A subset $A \subset X$ is said to be dense in X if

$$\overline{A} = X.$$

Example 1.2.45.1. As you know from your calculus course, the sets $\mathbb{Q}, \mathbb{Q}^c$ both are dense in $(\mathbb{R}, |\cdot|)$. But $\mathbb{R}$ is complete so it is not dense in $(\mathbb{C}, |\cdot|)$.

Example 1.2.45.2. This is a bit more non-trivial example. The set

$$G := \{m + \sqrt{2}n \ : \ m, n \in \mathbb{Z}\}$$

is dense in $(\mathbb{R}, |\cdot|)$.

Problem 1.2.46. Show that $\mathbb{Q}$ is not dense in $(\mathbb{R}, d)$ with the discrete metric.

Problem 1.2.47. If (X, d) is the discrete metric space then show that

$$\overline{A} = X$$

if and only if

$$A = X.$$

Example 1.2.47.1. This is a bit more non-trivial example. The set

$$S := \{\sin(n) \ : \ n \in \mathbb{Z}\}$$

is dense in $([-1, 1], |\cdot|)$.

Definition 1.2.48 (Equivalent Metric spaces)**.** Let (X, d_1) and (X, d_2) be two metrics on the same set X. We say that d_1 and d_2 are **equivalent** if there exist positive constants $C_1, C_2 > 0$ such that

$$C_1\, d_1(x, y) \leq d_2(x, y) \leq C_2\, d_1(x, y), \quad \forall x, y \in X.$$

Problem 1.2.49. Show that the above definition does not depend on the inequality that is: if there exist positive constants $c_1, c_2 > 0$ such that

$$c_1\, d_1(x, y) \leq d_2(x, y) \leq c_2\, d_1(x, y), \quad \forall x, y \in X.$$

if and only if there exist positive constants $a_1, a_2 > 0$ such that

$$a_1\, d_2(x, y) \leq d_1(x, y) \leq a_2\, d_2(x, y), \quad \forall x, y \in X.$$

Problem 1.2.50. Let (X, d_1) and (X, d_2) be a equivalent metric spaces. Then, (X, d_1) is complete if and only if (X, d_2) is complete.

Problem 1.2.51. Let (X, d), be a complete metric space. Let $M \subset X$ be a closed set then (M, d) is complete.

Example 1.2.51.1. Let $X = \mathbb{R}^n$. Define two metrics:

$$d_1(x, y) = \|x - y\|_2 = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}, \quad d_2(x, y) = \|x - y\|_1 = \sum_{i=1}^{n}|x_i - y_i|.$$

We claim that d_1 and d_2 are **equivalent metrics**. For any $x, y \in \mathbb{R}^n$:

(i) By the Cauchy-Schwarz inequality,

$$\|x-y\|_1 = \sum_{i=1}^{n} |x_i - y_i| \leq \sqrt{n}\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} = \sqrt{n}\,\|x-y\|_2.$$

(ii) Also, for each i, $|x_i - y_i|^2 \leq \sum_{j=1}^{n}(x_j - y_j)^2$, so

$$\|x-y\|_2 = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \leq \sum_{i=1}^{n} |x_i - y_i| = \|x-y\|_1.$$

Combining these inequalities gives

$$\|x-y\|_2 \leq \|x-y\|_1 \leq \sqrt{n}\,\|x-y\|_2, \quad \forall x, y \in \mathbb{R}^n.$$

Thus, taking constants $c = 1$ and $C = \sqrt{n}$, we have

$$c\, d_1(x,y) \leq d_2(x,y) \leq C\, d_1(x,y),$$

so d_1 and d_2 are equivalent metrics.

Example 1.2.51.2. Let $X = \mathbb{R}$, and consider the metrics:

$$d_1(x,y) = |x-y|, \quad d_2(x,y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

We claim that d_1 and d_2 are **not equivalent metrics**.

Suppose, the claim is not true. Then, there exist constants $c, C > 0$ such that

$$c\, d_1(x,y) \leq d_2(x,y) \leq C\, d_1(x,y), \quad \forall x, y \in \mathbb{R}.$$

Consider points $x, y \in \mathbb{R}$ with $x \neq y$ and $|x-y|$ very small, say $|x-y| = \varepsilon < \frac{1}{2C}$. Then

$$d_2(x,y) = 1 > C\, d_1(x,y) = C\varepsilon < \frac{1}{2} < 1,$$

which contradicts the upper inequality $d_2(x,y) \leq C d_1(x,y)$. Hence, no such constants $c, C > 0$ exist, and the metrics are not equivalent.

Proposition 1.2.52 (Completeness of the Bounded Metric)**.** The bounded metric on $\mathbb{R}$

$$d_b(x,y) = \frac{|x-y|}{1+|x-y|}, \quad \forall x, y \in X.$$

is also complete.

Proof. Let $\{x_n\}_{n\in\mathbb{N}}$ be a Cauchy sequence in (X, d_b). Since $f(t) = t/(1+t)$ is strictly increasing and continuous with inverse $f^{-1}(s) = s/(1-s)$, for $0 < \varepsilon < 1$ we have

$$d_b(x_m, x_n) < \varepsilon \implies |x_m - x_n| < f^{-1}(\varepsilon) = \frac{\varepsilon}{1-\varepsilon}.$$

Hence $\{x_n\}_{n\in\mathbb{N}}$ is a Cauchy sequence in $(\mathbb{R}, |\cdot|)$. By completeness of $(\mathbb{R}, |\cdot|)$, there exists $x \in X$ such that $x_n \to x$ in d. Finally, since d_b is continuous with respect to d, we also have $x_n \to x$ in (X, d_b). Therefore, (X, d_b) is complete. ∎

Problem 1.2.53. Show that on $\mathbb{R}$

$$x_n \xrightarrow{\frac{|\cdot|}{1+|\cdot|}} x$$

if and only if

$$x_n \xrightarrow{|\cdot|} x.$$

Problem 1.2.54. Show that the metric space $(\mathbb{R}, \dfrac{|\cdot|}{1+|\cdot|})$ and $(\mathbb{R}, |\cdot|)$ are not equivalent.

Problem 1.2.55. On $X = \{0,1\}^{\mathbb{N}} := \{f : \mathbb{N} \to \{0,1\}\}$ are the following metrics equivalent?

$$d_H(f,g) = \sum_{n=1}^{\infty} \frac{1}{2^n} |f(n) - g(n)|$$

The metric on this space is

$$d_C(f,g) := 2^{-N(f,g)} \text{ where } N(f,g) = \min_{n\in\mathbb{N}} |f(n) - g(n)| \neq 0 \text{ for } f \neq g;$$

and

$$d_C(f,f) := 0, \quad f \in X = \{0,1\}^{\mathbb{N}}.$$

Theorem 1.2.56. For a metric space (X, d) the following two statements are equivalent:

(i) (X, d) is complete.

(ii) For any descending sequence $A_1 \supset A_2 \supset A_3 \supset \cdots$ of nonempty closed sets with diameters approaching 0, then

$$\bigcap_{n\in\mathbb{N}} A_n \neq \emptyset.$$

Proof. (i) $\Rightarrow$ (ii): For each n, pick a point $a_n \in A_n$. Given $\varepsilon > 0$ we can find N large enough so that diam $(A_N) < \varepsilon$. Then, by using the descending condition,

$$d\,(a_n, a_m) < \epsilon \text{ for } n, m \geq N.$$

Hence the sequence $\{a_n\}_{n\in\mathbb{N}}$ is Cauchy. Since (X, d) is complete, $a_n \to a$ for some $a \in X$. We claim

$$a \in \bigcap_n A_n.$$

For take a fixed A_k. From a_k on, the points of the sequence $\{a_n\}$ lie in A_k. Since A_k is closed, $b \in A_k$. This happens for every $k \in \mathbb{N}$. Hence the claim follows.

Conversely for (ii) $\Rightarrow$ (i): Let $\{x_i\}$ be a Cauchy sequence in X; we have to prove that the sequence converges to a point in X. Define

$$B_i := \{x_i, x_{i+1}, \ldots\}.$$

Then we have $B_1 \supset B_2 \supset B_3 \supset \ldots$ and since $\{x_n\}$ is a Cauchy sequence,

$$\text{diam}(B_n) \to 0.$$

Define the closures

$$A_n = \overline{B_n}.$$

Then $A_1 \supset A_2 \supset A_3 \supset \cdots$ and $\text{diam}(A_n) = \text{diam}(\overline{B_n}) = \text{diam}(B_n)$, so

$$\text{diam}\,(A_i) \to 0.$$

By the hypothesis there exists $x \in \cap_n A_n$. And

$$d(x_n, x) \leq \text{diam}(A_n) \to 0.$$

So

$$x_n \to x.$$

■

Definition 1.2.57 (Open Cover and sub-cover)**.** Let (X, d) be a metric space. Let $A \subseteq X$. Then any collection of open sets $\{U_\alpha\}_{\alpha\in\Lambda}$ is called a cover for A if

$$A \subseteq \bigcup_{\alpha\in\Lambda} U_\alpha.$$

If $\Gamma \subseteq \Lambda$ then $\{U_\alpha\}_{\alpha\in\Lambda}$ is called a sub-cover if

$$A \subseteq \bigcup_{\alpha\in\Gamma} U_\alpha.$$

Definition 1.2.58 (Compact Metric space)**.** A metric space (X, d). A subset $A \subseteq X$ is called compact said to be compact if any open cover of A has finite subcover.

Remark 1.2.59. So, one can interpret the above definition as some sort of topological finiteness.

Proposition 1.2.60. Let (X, d) be a metric space and let $K \subseteq X$ be a compact subset. Then K is closed.

Proof. We will show that K^c is an open set. If $X = K$ then there is nothing to prove. So without loss of generality we can assume that $K \neq \emptyset$. Let $x \in K^c$. For any $x \in K$ there fix

$$\varepsilon_x = \frac{d(x, x_0)}{2},$$

Consider the open cover $\{B_{\varepsilon_x}(x)\}_{x\in K}$ of K, that is

$$K \subseteq \bigcup_{x\in K} B_{\varepsilon_x}(x).$$

Then there exists $x_1, x_2 \ldots, x_n$ such that

$$K \subseteq \bigcup_{i=1}^{n} B_{\varepsilon_{x_i}}(x_i).$$

If we fix

$$\varepsilon = \frac{1}{2} \min_{i\in\{1,\ldots n\}} (\varepsilon_i)$$

then

$$B_\varepsilon(x) \bigcap \left(\bigcup_{i=1}^{n} B_{\varepsilon_{x_i}}(x_i) \right) = \emptyset \Rightarrow B_\delta(x) \bigcap K = \emptyset.$$

In particular

$$B_\delta(x_0) \subseteq K^c.$$

So we conclude that K^c is an open set. ∎

Problem 1.2.61. Let (X, d) be a metric space $A \subseteq X$ be a finite set, then A is compact.

Theorem 1.2.62 (Finite intersection Properties)**.** Let (X, d) be a metric space. Then the following are equivalent.

1. Let $K \subseteq X$ be a compact set in (X, d).

2. For any collection $\{E_\alpha\}_{\alpha\in\Lambda}$ of closed subsets of K if

$$\bigcap_{\alpha\in\Lambda_0} E_\alpha \neq \emptyset \text{ for every finite subset } \Lambda_0 \text{ of } \Lambda \text{ then } \bigcap_{\alpha\in\Lambda} E_\alpha \neq \emptyset$$

Proof. Let K be a compact subset of (X, d). Suppose (ii) does not hold. That means there is a collection o $\{E_\alpha\}_{\alpha\in\Lambda}$ of closed subsets of K. Such that for any finite subset $\Lambda_0 \subseteq \Lambda$

$$\bigcap_{\alpha\in\Lambda_0} E_\alpha \neq \emptyset.$$

But

$$\bigcap_{\alpha\in\Lambda} E_\alpha = \emptyset \Rightarrow \bigcup_{\alpha\in\Lambda} E_\alpha^c = X,$$

hence

$$K \subseteq \bigcup_{\alpha\in\Lambda} E_\alpha^c.$$

Since, each E_α is closed E_α^c is open, in particular $\{E_\alpha\}_{\alpha\in\Lambda}$ is an open cover for K. Hence by compactness of K there exists a finite subset $\Lambda_0 \subseteq \Lambda$ such that

$$K \subseteq \bigcup_{\alpha\in\Lambda_0} E_\alpha^c$$

taking set complement on both sides we get

$$\bigcap_{\alpha\in\Lambda_0} E_\alpha \subseteq K^c.$$

But each $E_\alpha \subseteq K$, hence

$$\bigcap_{\alpha\in\Lambda_0} E_\alpha \subseteq K.$$

Notice, both the above containment hold only if and only if

$$\bigcap_{\alpha\in\Lambda_0} E_\alpha = \emptyset.$$

This contradicts the assumption that

$$\bigcap_{\alpha\in\Lambda_0} E_\alpha \neq \emptyset,$$

for any finite $\Lambda_0 \subseteq \Lambda$. In conclusion (ii) must hold.

$(ii) \Rightarrow (i)$

We prove this contradiction. Let $\{U_\alpha\}_{\alpha\in\Lambda}$ be an open cover of K. And suppose that K is not compact. That is, for every finite subset $\Lambda_0 \subseteq \Lambda$

$$K \nsubseteq \bigcup_{\alpha\in\Lambda_0} U_\alpha.$$

Consequently,using DeMorgan's laws we get

$$\emptyset \neq K \setminus \left(\bigcup_{\alpha \in \Lambda_0} U_\alpha \right) = \bigcap_{\alpha \in \Lambda_0} (K \setminus U_\alpha)$$

Note,

$$K \setminus U_\alpha = K \cap U_\alpha^c$$

is an intersection of closed sets and hence must be closed. So $\{(K \setminus U_\alpha)\}_{\alpha \in \Lambda}$ is a collection of closed sets contained in K. So by (ii)

$$\emptyset \neq \bigcap_{\alpha \in \Lambda} (K \setminus U_\alpha) .$$

By DeMorgan's laws

$$\emptyset \neq K \setminus \bigcup_{\alpha \in \Lambda} U_\alpha,$$

equivalently

$$K \nsubseteq \bigcup_{\alpha \in \Lambda} U_\alpha.$$

This contradicts our assumption that $\{U_\alpha\}$ is an open cover. Hence the supposition K is non-compact must be incorrect. And we conclude that K is a compact set. ∎

Proposition 1.2.63. Let (X, d) be a metric space and let $K \subseteq X$ be a compact set. Then $A \subseteq K$ is compact if and only if A is a closed in (X, d).

Proof. Let $\{U_\alpha\}_{\alpha \in \Lambda}$ be any arbitrary open cover of A. That is

$$A \subseteq \bigcup_{\alpha \in \Lambda} U_\alpha.$$

Since A is closed A^c is an open set. So

$$K \subseteq A^c \cup_{\alpha \in \Lambda} U_\alpha.$$

But by the hypothesis K is compact and hence there exists finite many $U_{\alpha_1}, U_{\alpha_2}, \cdots, U_{\alpha_n}$ such that

$$K \subseteq A^c \cup_{i=1}^n U_{\alpha_i},$$

note adding A^c does not harm. Moreover, as $A \cap A^c = \emptyset$,

$$A \subseteq \cup_{i=1}^n U_{\alpha_i}.$$

Hence for any arbitrary open cover of A we can find a finite sub-cover of A. So A is a compact set.

Conversely, if $A \subseteq K \subseteq X$ then A is close in (X, d) by Proposition 1.2.60. ∎

Proposition 1.2.64. Let (X, d) be a metric space and let K be compact subset in (X, d). Then K is complete.

Proof. Let $(x_n) \subseteq K$ be a Cauchy sequence. For each $n \in \mathbb{N}$ define the tail set

$$T_n := \{x_m : m \geq n\} \subseteq A$$

and let $\overline{T_n}$ denote its closure in X. Each $\overline{T_n}$ is nonempty (it contains T_n) and closed, and these sets are nested:

$$\overline{T_1} \supseteq \overline{T_2} \supseteq \overline{T_3} \supseteq \cdots .$$

We claim the family $\{\overline{T_n} : n \in \mathbb{N}\}$ has the finite intersection property. Indeed, for any finite subcollection $\overline{T_{n_1}}, \ldots, \overline{T_{n_k}}$ with $n_1 < \cdots < n_k$, we have $T_{n_k} \subseteq T_{n_j}$ for every $j \leq k$, so

$$\emptyset \neq T_{n_k} \subseteq \overline{T_{n_k}} \subseteq \bigcap_{j=1}^{k} \overline{T_{n_j}},$$

hence the finite intersection is nonempty. By Theorem1.2.62 $\overline{T_1} \subseteq A$), the intersection of all the closed sets is nonempty:

$$\bigcap_{n=1}^{\infty} \overline{T_n} \neq \emptyset.$$

Fix some $x \in \bigcap_{n=1}^{\infty} \overline{T_n}$ equivalently $x \in \overline{T_n}$ for every n. We claim that $x_n \xrightarrow{d} x$. Note, it follows from the claim that $\bigcap_{n=1}^{\infty} \overline{T_n}$ must be singleton.

Using the Cauchy property of $\{x_n\}_{n \in \mathbb{N}}$, given any $\varepsilon > 0$, choose N such that for all $p, q \geq N$,

$$d(x_p, x_q) < \frac{\varepsilon}{2}.$$

Since$x \in \bigcap_{n=1}^{\infty} \overline{T_n}$, we also have $x \in \overline{T_N}$. By using Theorem 1.2.32, there exists $m \geq N$ (i.e. $x_m \in T_N$) such that

$$d(x, x_m) < \frac{\varepsilon}{2}.$$

For any $n \geq N$ we then have

$$d(x_n, x) \leq d(x_n, x_m) + d(x_m, x) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

because $n, m \geq N$ and so $d(x_n, x_m) < \frac{\varepsilon}{2}$. This shows $x_n \to x$ as $n \to \infty$, and $x \in A$. Hence every Cauchy sequence in K converges to a point of K, so K is complete. ∎

Problem 1.2.65. Let $\{x_n\}_{n \in \mathbb{N}}$ be a Cauchy sequence on a metric space (X, d). If there is a subsequence $\{x_{n_k}\}_{k \in \mathbb{N}}$ of the above sequence such that

$$x_{n_k} \to x,$$

then show that

$$x_n \to x.$$

Problem 1.2.66. Given another proof of Proposition 1.2.64 using the above problem along with the nested sets argument.

Lemma 1.2.67. Let (X, d) be a metric space and let $K \subseteq X$ be a compact set. There there exists a countable set $Q \subseteq K$ such that

$$\overline{Q} = K$$

Proof. For each $n \geq 1$ consider the open covers $\{B_{1/n}(x)\}_{x \in K}$ of K. Then for every n there exists finitely many points $B_n := \{x_{n_1}, \dots, x_{n_r}\}$ such that

$$K \subseteq \bigcup_{i=1}^{r} B_{1/n}(x_{n_i})\}_{x \in K}.$$

We claim that

$$\overline{B} := \overline{\cup_{n \in \mathbb{N}} B_n} = K.$$

Since K is closed by Proposition 1.2.60 $\overline{B} \subseteq K$.

Let $x \in K$, for any given $\varepsilon > 0$ let $1/n, \varepsilon$. Then there exist a $B_{n_i}(x_{n_i})$ such that

$$x \in B_{1/n}(x_{n_i}) \to B_{1/n}(x) \cap B \neq \emptyset.$$

So $x \in \overline{B}$. ■

Theorem 1.2.68 (Sequential compactness criteria)**.** Let (X, d) be a metric space and let $K \subseteq X$. Then, following are equivalent.

(i) K is compact.

(ii) For any sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq A$ there exist a subsequence $\{x_{n_k}\}_{k \in \mathbb{N}}$ and a point $x \in A$ such that
$$\lim_{k \to \infty} x_{n_k} = x.$$

Proof. **(i) $\Rightarrow$ (ii).** Without loss of generality assume $K \neq \emptyset$., and $\{x_n \;:\; n \in \mathbb{N}\} \subseteq K$ is an infinite set.

Consider, the open balls $\{B_1(x) : x \in K\}$, then

$$K \subseteq \bigcup_{x \in K} B_1(x).$$

By compactness, there exists finitely many points a_1, a_2, a_n of K such that

$$K \subseteq \bigcup_{i=1}^{n} B_1(a_i).$$

So at least one of above balls will contain infinite any points of the sequence. Say $B_1 := B_1(a_0)$, contains infinitely many terms of the sequence. Pick a subsequence $\{x^1_{n_k}\}_{k\in\mathbb{N}}$ consisting of those terms lying in B_1. Cover $\overline{B_1}$ by balls of radius 1/2,

$$\overline{B_1} \subseteq \bigcup_{x\in\overline{B_1}} B_{\frac{1}{2}}(x).$$

By Proposition 1.2.60, the set $\overline{B_1}$ is also compact. Then

$$\overline{B_1} \subseteq \bigcup_{x\in\overline{B_1}} B_{1/2}(x).,$$

and hence there exits some finitely many $a_1, \dots, a_n$ such that

$$\overline{B_1} \subseteq \bigcup_{i=1}^{n} B_{1/2}(a_i).$$

Hence, after possible renaming, $B_{1/2}(a_1)$ contains infinitely many terms of the sequence $\{x^{(1)}_{n_k}\}_{k\in\mathbb{N}}$. Call those term as $\{x^{(2)}_{n_k}\}$. Then

Continue inductively to obtain a nested sequence of balls

$$\overline{B_1} \supseteq \overline{B_2} \supseteq \overline{B_3} \supseteq \cdots \tag{2.4}$$

with $\operatorname{diam}(B_r) = 2^{r-1}$. And each B_k contains infinitely many terms of the corresponding subsequence $\{x^{(k)}_n\}_{n\in\mathbb{N}}$ of the sequence $\{x_n\}_{n\in\mathbb{N}}$.

Choose a diagonal subsequence by picking $x_{n_p} = x^{(p)}_{n_p} \in B_p$ with indices strictly increasing in p. Then for $m \geq r \geq N$ we have

$$d(x_{n_m}, x_{n_r}) \leq \operatorname{diam}(B_r) = 2^{r-1},$$

Hence $\{x_{n_k}\}_{k\in\mathbb{N}}$ is Cauchy, in the compact set A, and by Proposition **??** A is complete. So Cauchy subsequence converges to some limit $x \in A$.

(ii) ⇒ (i). Assume every sequence in K has a convergent subsequence whose limit lies in K. We show K is compact.

Let $\mathcal{U} := \{U_i\}_{i\in I}$ be an open cover of K. We first claim that there exist a $\delta > 0$ such that for every $x \in K$ there is a $U_i \in \mathcal{U}$ (depending on x) such that

$$x_{\in} B_\delta(x) \subseteq U_i.$$

Note, the δ above is independent of any $x \in K$.

If this claim is not true then for every $n \in \mathbb{N}$ there exist $x_n \in K$ such that

$$B_{\frac{1}{n+1}}(x_n) \nsubseteq U_i, \quad \forall i \in I. \tag{2.5}$$

By sequential compact there is subsequence $\{x_{n_k}\}$ of $\{x_n\}$ such that x_{n_k} is convergent to some point $x_0 \in K$.

But, since $\mathcal{U}$ is a open cover of K, there exist a U_{i_0} such that $x_0 \in U_{i_0}$. And since U_{i_0} is open there exist $\varepsilon > 0$ such that

$$B_\varepsilon(x_0) \subseteq U_{i_0}.$$

By definition of convergence, for this $\varepsilon > 0$ there is $N > 0$, such that

$$x_{n_k} \in B_\varepsilon(x_0), \quad \forall k > N.$$

Moreover, there exist a $k_0 > N$ large enough such that

$$\frac{1}{n_{k_0} + 1} < \frac{\varepsilon}{2}.$$

Then for that fixed $x_{n_{k_0}} \in B_\varepsilon(x_0) \subseteq U_{i_0}$; we have

$$B_{\frac{1}{n_{k_0}+1}}(x_{n_{k_0}}) \in B_\varepsilon(x_0) \subseteq U_{i_0}.$$

This gives a contradiction to eq 2.5. Hence the claim must hold so, there exist $\delta > 0$ such that for any $x \in K$, there exist U_i (depending on x) and

$$B_\delta(x) \subseteq U_i.$$

Now suppose, that K is not compact. Let δ be fixed as above. Without loss of generality assume that the open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of K with no finite sub-cover.

Pick $x_1 \in K$ some fixed point. By then there exist U_{i_1} such that

$$x_1 \in B_\delta(x_i) \subseteq U_{i_1}.$$

Since $\{U_i\}_{i \in I}$ doesn't admit a finite sub-cover,

$$K \not\subseteq U_{i_1},$$

there exists

$$x_2 \in K \setminus U_{i_1}.$$

But there exists U_{i_2} such that $x_2 \in U_{i_2}$, and

$$x_1 \in B_\delta(x_i) \subseteq U_{i_2}.$$

Since, $\mathcal{U}$ does not possess any finite sub-cover

$$K \not\subseteq U_{i_1} \cup U_{i_2},$$

there exist $x_3 \in K \setminus U_{i_1} \cup U_{i_2}$. And there exist U_{i_3} containing x_3 such that

$$x_3 \in B_\delta(x_3) \subseteq U_{i_3}.$$

Keep on choosing, inductively, we get choose $x_n \in A \setminus \bigcup_{j=1}^{n-1} U_{i_j}$. There exists an open set U_{i_n} such that

$$x_n \in B_\delta(x_n) \subseteq U_{i_n}.$$

We leave it as a quick verification to show that $\{x_n\}_{n\in\mathbb{N}}$ does not even have any Cauchy subsequence, as each consecutive ter is 2δ distance apart.

Combining the two implications, (i) and (ii) are equivalent. ■

Remark 1.2.69. The following fact is part of the above proof: Let $K \subseteq X$ is sequentially compact in (X, d). Then for any given open cover $\{U_i\}_{i\in I}$ of K, there exists a $\delta > 0$ such that for any $x \in K$ there exist $U_i, \quad i \in I$ (depending on x) with

$$B_\delta(x) \subseteq U_i.$$

The δ is called the **Lebesgue Number** of the cover $\{U_i\}_{i\in I}$ for the compact set K.

Problem 1.2.70. Show that in a be discrete metric space (X, d), a subset $K \subseteq X$ is a compact set if and only if K is finite.

Problem 1.2.71 (Bolzano Weierstrass Theorem)**.** Let $\{x_n\}_{n\in\mathbb{N}}$ be abounded sequence in $(\mathbb{R}, |\cdot|)$, then there exist a subsequence $\{x_{n_k}\}_{k\in\mathbb{N}}$ and a point $x \in \mathbb{R}$ such that

$$x_{n_k} \xrightarrow{|\cdot|} x.$$

Hint: Since $\{x_n\}_{n\in\mathbb{N}}$ is given to be a bounded sequence there exist $M > 0$ such that we can take

$$\{x_n\}_{n\in\mathbb{N}} \subseteq [-M, M].$$

Now Use the idea of dividing the set $[-M, M]$ into partitions, as in the first part of sequential compactness, to construct a Cauchy subsequence of $\{x_n\}$. Conclude the result by completeness of $\mathbb{R}$.

Theorem 1.2.72 (Hiene Borel theorem for $\mathbb{R}$.)**.** A subset $K \subseteq \mathbb{R}$ is compact in $(\mathbb{R}, |\cdot|)$ if and only if K is closed and $\mathrm{diam}(K) < \infty$, (i.e K is bounded).

Proof. The proof is left as an exercise, follow th hints below. (i)⇒(ii) is true for amy compact subset of any metric space and has been proved before.

For (ii)⇒(i) First use Bolzano Weierstrass Theorem and then use the fact that K is closed. ■

Remark 1.2.73. A more general version of Heine-Borel theorem is as follows: A subset $K \subset \mathbb{C}^n$ is compact with respect any norm if and only K is closed and bounded in the Euclidean norm.

Problem 1.2.74. Let $M \subseteq \mathbb{R}$ be a closed set in $(\mathbb{R}, |\cdot|)$. If $\inf(M) \in \mathbb{R}$ then show that $\inf(M) \in M$. In particular if $K \subseteq \mathbb{R}$ is a compact subset in $(\mathbb{R}, |\cdot|)$ then show that

$$\inf(K), \sup(K) \in K.$$

Problem 1.2.75. Using the above problem show that if $K \subseteq X$ be a compact set in (X, d) and $fK \to \mathbb{R}$ is a continuous function then show that there exists $x_0, y_0 \in K$ such that

$$f(x_0) = \inf_{x \in K} f(x_0), \quad f(y_0) = \sup_{x \in K} f(x_0),$$

Problem 1.2.76. Let (X, d_1) and (X, d_2) be two equivalent metric spaces. Then show that any $K \subseteq X$ is compact in (X, d_1) if and only if K is compact in (X, d_2).

The following part is additional and it's not covered in the class. If we cover this in class then I will expand this part.

Definition 1.2.77 (Connected Metric space)**.** A metric space (X, d) is said to be disconnected if X contains a non-trivial subset A which is both open and closed in (X, d). That is, A is both open and closed in (X, d) as $A \neq \emptyset$ and $A \neq X$.

A metric space is said to be connected if it is not disconnected.

A subset of is said to be connected/disconnected if the metric subspace is connected/disconnected.

Theorem 1.2.78 (Connectedness Criterion)**.** Let (X, d) be a metric space, then the following are equivalent.

1. (X, d) is connected.
2. There exists open sets $U, V \subseteq X$ such that

$$X = U \cup V \text{ and } U \cap V = \emptyset.$$

Example 1.2.78.1. $(\mathbb{R}, |\cdot|)$ is connected metric space.

1.3 Functions on Metric spaces

Definition 1.3.1 (Continuity $\varepsilon - \delta$ definition)**.** Let (X_1, d_1) and (X_2, d_2) be two metric spaces, let $f : X_1 \to X_2$ be any function. Let $x_0 \in X_1$ be any point. We say that f is continuous at x_0 if the for any $\varepsilon > 0$ there exists $\delta > 0$ (possibly depending on x_0) such that

$$d_1(x, x_o) < \delta \Rightarrow d_2(f(x), f(x_0)) < \epsilon.$$

Moreover, we say that f is continuous on X_1 if f is continuous on every $x \in X_1$.

The following notations are convenient. Let (X, d_1) and (Y, d_2) be two metric spaces, let $f : X \to Y$ then

$$f(A) = \{f(x) \;:\; x \in A\}, \quad A \subseteq X_1.$$

Note, if $A = \emptyset$ then $f(A) = \emptyset$. The set $f(A)$ is called the image of f on A. And $f(X)$ is simply called the image of f.

$$d_1\left(x, x_o\right) < \delta d_2\left(f(x), f(x_0)\right) < \epsilon. \tag{3.6}$$

But

$$\{x \;:\; d_1\left(x, x_o\right) < \delta\} = B_\delta(x_0),$$

and

$$\{z \;:\; d_2\left(f(x), f(x_0)\right) < \epsilon\} = B_\varepsilon(f(x_0)).$$

So we can rewrite eq 3.6

$$x \in B_\delta(x_0) \Rightarrow f(x) \in B_\varepsilon(f(x_0)).$$

In particular, if f is continuous at x_0 then for any given $\varepsilon > 0$ there exist a $\delta > 0$ (depending on ε) such that

$$f(B_\delta(x_0)) \subset B_\varepsilon(f(x_0)).$$

We can characterize continuity at a point in terms of neighborhoods, or in terms of convergence. The characterization in terms of neighborhoods is just a slight change of language.

Theorem 1.3.2. Let $(X_1, d_1), (X_2, d_2)$ be two metric spaces. And let $f : X_1 \longrightarrow X_2$ be a function. Then the following are equivalent:

(i) f is continuous at $x_0 \in X$.

(ii) If $x_n \xrightarrow{d_1} x_0$ then $f(x_n) \xrightarrow{d_2} f(x_0)$.

Proof. $(i) \Rightarrow (ii)$

Let f be continuous at x_0. Let $x_n \xrightarrow{d_1} x_0$. Let $\varepsilon > 0$ be arbitrarily chosen, them there exist $\delta > 0$ such that

$$f(B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0)).$$

For this $\delta > 0$ there exist $N \in \mathbb{N}$ such that

$$d(x_n, x_0) < \varepsilon \Rightarrow x_n \in B_\delta(x_0), \quad n \geq N.$$

So for $\varepsilon > 0$ there exists the above $N \in \mathbb{N}$ such that

$$f(x_n) \in B_\varepsilon(f(x_0)) \Rightarrow d_2(f(x_n), f(x_0)) < \varepsilon, \quad n \geq N.$$

Hence $f(x_n) \xrightarrow{d_2} f(x_0)$.

$(ii) \Rightarrow (i)$

We prove the contrapositive. Suppose f is not continuous at x_0. Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ there exists $x \in X_1$ satisfying

$$d_1(x, x_0) < \delta \quad \text{but} \quad d_2(f(x), f(x_0)) \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, choose $x_n \in X_1$ with

$$d_1(x_n, x_0) < \frac{1}{n} \quad \text{and} \quad d_2(f(x_n), f(x_0)) \geq \varepsilon_0.$$

Then $x_n \to x_0$ in (X_1, d_1), but $f(x_n)$ does not converge to $f(x_0)$ in (X_2, d_2), contradicting (ii). Hence f must be continuous at x_0. ■

Corollary 1.3.2.1. Let $(X_1, d_1), (X_2, d_2)$ and (X_3, d_3) be three metric spaces. Let $g : X_1 \to X_2$ and $f : X_2 \to X_3$ be continuous functions. Then

$$f \circ g : X_1 \to X_3 \text{ is a continuous function.}$$

Proof. Use sequential criterion of continuity. ■

Corollary 1.3.2.2. Let (X_1, d_1) and (X_2, d_2) be metric spaces, $f : X_1 \to X_2$ continuous, and $Y \subseteq X_1$. Then the restriction $f|_Y : (Y, d_1) \to (X_2, d_2)$ is also a continuous function.

Proof. Let $y \in Y$ and let $\{y_n\}_{n\in\mathbb{N}} \subset Y$ be any sequence with $y_n \xrightarrow{d_1} y$ in Y. Convergence in the subspace metric is equivalent to convergence in the ambient space, so $y_n \to y_0$ in (X_1, d_1).

Since f is continuous on X_1, it preserves limits of sequences, hence

$$f(y_n) \xrightarrow{d_2} f(y_0) \quad \text{in } (X_2, d_2).$$

As this holds for every sequence in Y converging to y_0, the restriction $f|_Y$ is continuous at y_0. Since y_0 was arbitrary, f is continuous on Y. ■

Definition 1.3.3 (Image and Inverse image of a function)**.** Let X_1, X_2 be any two sets and let $f : X_1 \to X_2$ be any function. For set $B \subseteq X_2$ we define inverse image of the function as

$$f^{-1}(B) = \{x \in X_1 \ : \ f(x) \in B\}$$

For any set $A \subseteq X_1$ we define image of f as

$$f(A) = \{y \in X_2 \ : \ \exists x \in A \ni y = f(x)\}.$$

Definition 1.3.4 (Image and Inverse Image)**.** Let $f : X \to Y$ be a function. For subsets $A, B \subseteq X$ and $C, D \subseteq Y$, define:

$$f(A) = \{f(x) : x \in A\} \subseteq Y, \quad f^{-1}(C) = \{x \in X : f(x) \in C\} \subseteq X.$$

Properties of Image and Inverse Image Let $f : X \to Y$ be a function. Then the following hold.

1 **Properties of Image:**

i (*Monotonicity*) If $A \subseteq B \subseteq X$, then
$$f(A) \subseteq f(B).$$

ii (*Union*)
$$f(A \cup B) = f(A) \cup f(B).$$

iii (*Intersection*)
$$f(A \cap B) \subseteq f(A) \cap f(B),$$
with equality if f is injective.

iv (*Difference*)
$$f(A \setminus B) \subseteq f(A) \setminus f(B),$$
with equality if f is injective.

v (*Special sets*)
$$f(\emptyset) = \emptyset, \qquad f(X) = \text{range}(f).$$

2 **Properties of Inverse Image:**

i (*Monotonicity*) If $C \subseteq D \subseteq Y$, then
$$f^{-1}(C) \subseteq f^{-1}(D).$$

ii (*Union*)
$$f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D).$$

iii (*Intersection*)
$$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D).$$

iv (*Complement*)
$$f^{-1}(Y \setminus C) = X \setminus f^{-1}(C).$$

v (*Special sets*)
$$f^{-1}(\emptyset) = \emptyset, \qquad f^{-1}(Y) = X.$$

3 **Relations between Image and Inverse Image:**

i For every $A \subseteq X$,
$$A \subseteq f^{-1}(f(A)),$$
with equality if f is injective.

ii For every $C \subseteq Y$,
$$f(f^{-1}(C)) \subseteq C,$$
with equality if f is surjective.

iii (*Composition*) For $g : Y \to Z$ and $E \subseteq Z$,
$$(g \circ f)^{-1}(E) = f^{-1}(g^{-1}(E)).$$

Proposition 1.3.5. Let Let $(X_1, d_1), (X_2, d_2)$ be two metric spaces. And let $f : X_1 \longrightarrow X_2$ be a function. Then following statements are equivalent

(i) f is continuous on X_1.

(ii) If $V \subseteq X_2$ is an open set in (X_2, d_2) then $f^{-1}(V) = \{x \in X_1 \ : \ f(x) \in V\}$ is open in (X_1, d_1).

Proof. $(ii) \Rightarrow (i)$ follows from definition of continuity on X_1.

For $(i) \Rightarrow (ii)$ consider any open set $V \subseteq X_2$. If $Y = \emptyset$ then there is nothing to prove as $f^{-1}(\emptyset) = \emptyset$ which is trivially open. So assume that $V \neq \emptyset$. Let $x \in f^{-1}(V)$ be any arbitrary point. So there exits a $y \in V$, such that $f(x) = y$. Moreover, as V is assumed to be open there exist $\varepsilon > 0$ such that $B_\varepsilon(y) \subseteq V$. Since f is assumed to be continuous on X_1, in particular, f is continuous at x. So For the above ε there exists a $\delta > 0$ (depending on ε and x) such that
$$f(B_\delta(x)) \subseteq B_\varepsilon(y) \Rightarrow B_\delta(x) \subseteq f^{-1}(B_\varepsilon(y)).$$
But
$$B_\varepsilon(y) \subseteq V \Rightarrow f^{-1}(B_\varepsilon(y)) \subseteq f^{-1}(V).$$
Hence
$$B_\delta(x) \subseteq f^{-1}(V).$$
So, for any $x \in f^{-1}(V)$ arbitrary point there exist a $\delta > 0$ such that
$$B_\delta(x) \subseteq f^{-1}(V).$$
In conclusion $f^{-1}(V)$ is an open set. ■

Definition 1.3.6 (Isometry)**.** A function f between two metric spaces (X_1, d_1) and (X_2, d_2) is said to be an isometry if it preserves distance. That is $f : X_1 \to X_2$ is an isometry if
$$d_2(f(a), f(b)) = d_1(a, b) \text{ for all} a, b \in X_1.$$

Problem 1.3.7. Show that every isometry is a continuous function.

Problem 1.3.8. If (X, d) be a discrete metric space and let (Y, d') be any metric space then any function $f : X \to Y$ is continuous.

Definition 1.3.9 (Lipschitz map)**.** A function f between two metric spaces (X_1, d_1) and (X_2, d_2) is said to be an Lipschitz function if there exists a $k \geq 0$ such that

$$d_2(f(a), f(b)) \leq k \cdot d_1(a, b).$$

The least of such k's is called the Lipschitz-constant of f. If Lipschitz-constant $k < 1$ then f is called a contraction.

Remark 1.3.10. Observe if $f : X_1 \to X_2$ is a Lipschitz continuous function on metric space (X_1, d_1) to (X_2, d_2) with Lipschitz constant $k > 0$ then fro any $x \neq y \in X_1$

$$\frac{d_2(f(x), f(y))}{d_1(x, y)} \leq k.$$

Problem 1.3.11. Show that every Lipschitz map is continuous.

Example 1.3.11.1. Let (X, d_1) and (X, d_2) be two equivalent metric spaces on the set X. By definition of equivalent matrix, there exists $\alpha, \beta > 0$ such that

$$\alpha\ d_2(x, y) \leq d_1(x, y) \leq \beta d_2(x, y)$$

Hence the identity map $I : X \to X$ is a Lipschitz map from the metric space (X, d_1) to (X, d_2) with Lipschitz constant β. As well as, I is a Lipschitz map from the metric space (X, d_2) to (X, d_1) with Lipschitz constant α.

Example 1.3.11.2. Let f be a function from $([0, 1], |\cdot|)$ to $(\mathbb{R}, |\cdot|)$ defined as

$$f(x) = x^2.$$

Then, for any $x, y \in [0, 1]$

$$|f(x) - f(y)| = |x^2 - y^2| = |(x - y)||(x + y)|$$

But $x, y \in [0, 1]$ and hence

$$|x + y| \leq 2.$$

So we get

$$|f(x) - f(y)| = |x^2 - y^2| = |(x - y)||(x + y)| \leq 2|x - y|.$$

Hence $f(x) = x^2$ is a Lipschitz map from $[0, 1]$.

Problem 1.3.12. Let (X, d) be a metric spaces. Let $f : X \to X$ be a continuous function. Then Show that the following map is Lipschitz continuous function from X to $\mathbb{R}$

$$x \mapsto d(x, f(x)).$$

Proposition 1.3.13. Let (X_1, d_1) and (X_2, d_2) be metric spaces. Let $f : X_1 \to X_2$ be a continuous function. Let $D \subseteq X_1$ then

$$\overline{D} = X \Rightarrow \overline{f(D)} = f(X).$$

Proof. Let (X_1, d_1) and (X_2, d_2) be metric spaces. Let $f : X_1 \to X_2$ be a continuous function. Let $D \subseteq X_1$ be such that

$$\overline{D} = X.$$

Fix some $f(x) \in f(X) \subset X_2$. So, by Theorem 1.2.32 for every $x \in X_1$ there exists $\{x_n\}_{n\in\mathbb{N}} \subset D$ be sequence such that $x_n \xrightarrow{d_1} x$.

Then by Theorem 1.3.2

$$f(x_n) \xrightarrow{d_2} f(x).$$

Hence replying Theorem 1.2.32 we get

$$\overline{f(D)} = f(X).$$

■

The following problem problem is a bit tough, nevertheless you can try.

Problem 1.3.14. Using continuous function preserve density, Show that the set

$$S := \{\sin(n) \ : \ n \in \mathbb{N}\}$$

is dense in $([-1, 1], |\cdot|)$.

Hint: Consider the metric space $([0, 2\pi), |\cdot|)$ as the subspace of $(\mathbb{R}, |\cdot|)$.

Then show that

(i) $\sin : [0, 2\pi) \to [-1, 1]$ is a continuous function.

(ii) Show that

$$\{[n]_{2\pi} := n - k2\pi$$

Proposition 1.3.15. Let $(X_1, d_1), (X_2, d_2)$ be two metric spaces. Let $f : X_1 \to X_2$ be continuous function. Then: If $K \subseteq X_1$ be a compact subset then $f(K)$ is a compact subset of X_2.

Proof. Left as an exercise. Hint: Use sequential Compactness criteria. ■

Definition 1.3.16 (Uniform Continuity)**.** Let (X_1, d_1) and (X_2, d_2) be two metric spaces. Let $f : X_1 \to X_2$ be a function. Then f is said to be uniformly continuous on X_1 if for any $\varepsilon > 0$ there exist $\delta > 0$ such that

$$f(B_\delta(x)) \subseteq B_\varepsilon(f(x)), \quad \forall x \in X_1.$$

Problem 1.3.17. Show that if $f : X_1 \to X_2$ is Lipschtiz then f is uniformly continuous.

Theorem 1.3.18. Let (X_1, d_1) and (X_2, d_2) be metric spaces, and let $f : X_1 \to X_2$ be a uniformly function f preserves Cauchy sequences, i.e. for every Cauchy sequence (x_n) in X_1, the sequence $(f(x_n))$ is Cauchy in X_2.

Proof. Suppose f is uniformly continuous. Then

$$\forall \varepsilon > 0,\ \exists \delta > 0 \text{ such that } d_1(x, y) < \delta \Rightarrow d_2(f(x), f(y)) < \varepsilon.$$

Let (x_n) be a Cauchy sequence in X_1. Then there exists N such that for all $m, n \geq N$, $d_1(x_m, x_n) < \delta$. By uniform continuity,

$$d_2(f(x_m), f(x_n)) < \varepsilon.$$

Thus $(f(x_n))$ is Cauchy in X_2. ∎

Problem 1.3.19. Find an example to show that the converse of the above theorem is false in general.

Theorem 1.3.20. Let $(X, d_1), (X_2, d_2)$ be metric spaces and let (X, d_1) be a compact. If $f : X_1 \to X_2$ be continuous on X_1 then f is uniformly continuous.

Proof. We will prove by contradiction. Suppose that f is not uniformly continuous. Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ we can find points $x_\delta, y_\delta \in X$ satisfying

$$d_X(x_\delta, y_\delta) < \delta \quad \text{but} \quad d_2\big(f(x_\delta), f(y_\delta)\big) \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, take $\delta_n = \frac{1}{n}$ and define $x_n = x_{\delta_n}$ and $y_n = y_{\delta_n}$. Then

$$d_1(x_n, y_n) < \frac{1}{n} \quad \text{and} \quad d_2\big(f(x_n), f(y_n)\big) \geq \varepsilon_0 \quad \text{for all } n \in \mathbb{N}.$$

Since X_1 is compact and equivalently (by Theorem **??**) X_1 is sequentially compact, the sequence $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$ such that $x_{n_k} \to x$ for some $x \in X$. Moreover, because $d_X(x_{n_k}, y_{n_k}) \leq \frac{1}{n_k} \to 0$, we also have $y_{n_k} \to x$.

By the continuity of f, we have

$$f(x_{n_k}) \to f(x) \quad \text{and} \quad f(y_{n_k}) \to f(x).$$

Hence,

$$d_Y\big(f(x_{n_k}), f(y_{n_k})\big) \to 0.$$

This contradicts the fact that $d_Y(f(x_{n_k}), f(y_{n_k})) \geq \varepsilon_0$ for all k. Therefore, our assumption was false, and f must be uniformly continuous. ∎

Example 1.3.20.1. Consider the function $f : [0,1] \to \mathbb{R}$ defined by $f(x) = \sqrt{x}$.

Since f is continuous on the closed and bounded interval $[0,1]$, it is uniformly continuous. Alternatively, for all $x, y \in [0,1]$ with $x \neq y$,

$$|\sqrt{x} - \sqrt{y}| = \frac{|x-y|}{\sqrt{x} + \sqrt{y}} \leq |x-y|^{1/2}.$$

Hence f is uniformly continuous on $[0,1]$.

We claim that f is not Lipschitz. Suppose there exists $L > 0$ such that

$$|\sqrt{x} - \sqrt{y}| \leq L|x-y| \quad \forall\, x, y \in [0,1].$$

Taking $x = 0$ and $y = t > 0$, we get

$$\frac{|\sqrt{t} - \sqrt{0}|}{|t-0|} = \frac{\sqrt{t}}{t} = \frac{1}{\sqrt{t}} \to \infty \quad \text{as } t \to 0^+.$$

Hence, no such finite L exists, and f is not Lipschitz on $[0,1]$.

Therefore, $f(x) = \sqrt{x}$ is uniformly continuous on $[0,1]$ but not Lipchitz.

Remark 1.3.21. In general we have the following inclusion of function

$$\text{Lipschitz Continuous} \Rightarrow \text{Uniformly Continuous} \Rightarrow \text{Continuous}$$

An we saw an example of Uniformly Continuous function that is on Lipschitz. There is also continuous function that is not uniformly continuous. Consider the euclidean subspace $((0,1), |\cdot|)$. An the function

$$f : (0,1) \Rightarrow \mathbb{R} \quad f(x) = \frac{1}{x}.$$

Then, considering the euclidean metric on the co-domain space $\mathbb{R}$, the above function is continuous of $(0,1)$. But $f(x) = \dfrac{1}{x}$ is not uniformly continuous. Indeed, the image of the Cauchy sequence $\left\{\dfrac{1}{n}\right\}_{n\in\mathbb{N}}$ under f is $\{n\}_{n\in\mathbb{N}}$ which is clearly not Cauchy.

Problem 1.3.22. Let $f : K \to X_2$ be continuous function. If $K \subseteq X_1$ be a compact set in (X_1, d) then, using the sequential compactness, show that $f(K)$ is a compact set in $(X_2, d_2.)$

Problem 1.3.23. Let (X_1, d_1) and (X_2, d_2) be metric spaces. Let $f : X_1 \to X_2$, be continuous function such that $f(X_1)$ is compact subset of X_2. Does this necessarily imply that f is uniformly continuous.

1.3.1 Fixed Point Theorem

A fixed-point theorem is an assertion of the following kind: For a suitable mapping f on a suitable set X there exists a point $x \in X$ with $f(x) = x$. For metric spaces there is a very simple fixed-point theorem for mappings which might appropriately be called strict contractions.

Theorem 1.3.24 (Brouwer Fixed point theorem)**.** Let $\mathbb{D} \subseteq \mathbb{R}^2$ the closed unit disc, and if $f : \mathbb{D} \to \mathbb{D}$ be continuous function, then there exist $x \in \mathbb{D}$ such that

$$f(x) = x.$$

Theorem 1.3.25 (Edelstein's fixed point)**.** Let (X, d) be a metric space. And let $M \subseteq X$ be a compact metric space. Let $f : M \to X$ be a function on M such that

$$d(f(x), f(y)) < d(x, y), \quad x \neq y,$$

(Such a function is called a strict contraction). Then there exist a $x_0 \in M$ such that a $f(x_0) = x_0$.

Proof. Suppose (X, d) be a metric space. And $M \subseteq X$ be a compact metric space. Let $f : M \to X$ be any continuous function on M such that

$$d(f(x), f(y)) < d(x, y), \quad x \neq y.$$

The define

$$h(x) = d(x, f(x)).$$

We claim that $h(x)$ is Lipschtiz (continuous) function with respect to the euclidean metric on $\mathbb{R}$. Let $x \neq y \in X$ then

$$h(x) = d(x, f(x)) \leq d(x, y) + d(y, f(y)) + d(f(x), f(y))$$

consequently, using the fact that f is strict contraction;

$$h(x) - h(y) < d(x, y) + d(x, y) = 2d(x, y)$$

Hence

$$|h(x) - h(y)| < 2d(x, y), \quad x \neq y \in M.$$

So h is a Lipschtiz continuous function. Since $h : M \to \mathbb{R}$ is a continuous function the image $f(M)$ of M under of f is compact. By Hiene-Borel theorem every compact subset of $\mathbb{R}$ is closed and bounded. Hence $0 \leq |h(x)| \leq M$. We claim there exist $x_0 \in M$ such $h(x_0) = 0$.

There exits a $x_0 \in M$ such that

$$h(x_0) = \inf_{x \in M} |h(x)|.$$

If not then
$$h(f(x_0)) = d(f(x_0), f(f(x_0))) < d(x_0, f(x_0)) = h(x_0),$$
this contradicts the fact that $h(x_0) = \inf_{x\in M} |h(x)|$.

Hence $h(x_0) = 0$ in particular

$$h(x_0) = d(x_0, f(x_0)) = 0 \Rightarrow f(x_0) = x_0.$$

■

Theorem 1.3.26 (Banach Fixed Point or Contraction Mapping)**.** Let (X, d) be a complete metric space and let $f : X \to X$ be a mapping with the following property: There exists a real number $0 < k < 1$ such that

$$d(f(x), f(y)) < kd(x, y)$$

for all $x \neq y \in X$. Then there exist unique $x \in X$ such that

$$f(x) = x.$$

Proof. The uniqueness is immediate for if $f(x) = x$ and $f(y) = y$ with $x \neq y$, then

$$d(f(x), f(y)) = d(x, y),$$

contrary to the hypothesis.

To prove the existence of a fixed point, we construct a Cauchy sequence by iterating f. Fix any point $z \in X$. If $z = f(z)$ then there is nothing to do, so without loss of generality assume that $d(z, f(z)) \neq 0$. Now we consider the sequence

$$\{z, f(z), \cdots, f^{(n)}(z), \cdots\}_{n\in\mathbb{N}},$$

where
$$f^n(z) = f \circ f \circ \cdots \circ f(z)\text{composition of } f \text{ } n\text{-times}.$$

Note, by hypothesis on the function,

$$d\left(f^n(z), f^{n-1}(z)\right) \leq k^{(n-1)} d\left(f(z), z\right). \tag{3.7}$$

Hence, by iterating and using the triangle inequality, we have, for any $m > n$,

$$\begin{aligned} d\left(f^m(z), f^n(z)\right) &\leqq d\left(f^m(z), f^{m-1}(z)\right) + d\left(f^{m-1}(z), f^{m-2}(z)\right) + \cdots + d\left(f^{n+1}(z), f^n(z)\right) \\ &\overset{3.7}{\leqq} \left(k^{m-1} + k^{m-2} + \cdots + k^n\right) d\left(f(z), z\right) \\ &= \left(\sum_{i=1}^{m-1} k^i - \sum_{i=1}^{n-1} k^i\right) d(f(z), z). \end{aligned} \tag{3.8}$$

But, since $0 < k < 1$, the series $s_n := \sum_{i=1}^{n} k^i$ is convergent and hence in particular it is Cauchy. And hence for any $\varepsilon > 0$ there exist a $N \in \mathbb{N}$ such that

$$s_{m-1} - s_{n-1} < \varepsilon \quad \text{for every} m, n > N.$$

Consequently, for $\varepsilon > 0$, and $N \in \mathbb{N}$ as above, using eq 3.8 we get

$$d\left(f^m(z), f^n(z)\right) \leqq (s_{m-1} - s_{n-1}) d(z, f(z)) \leqq \varepsilon d(z, f(z)) \leqq \epsilon, \quad \forall m > n > N.$$

Now since, $d(z, (f(z)) > 0$ is fixed, we have concluded that $\{f^{(n)}(z)\}_{n \in \mathbb{N}}$ is a Cauchy sequence.

Finally (X, d) is assumed to be complete. So, there exists $w \in X$ such that

$$f^n(z) \xrightarrow{d} w.$$

We show that $f(w) = w$. Indeed,

$$f(w) = f(\lim_n f^n(z)),$$

but f is continuous on X, in particular f is continuous at w so

$$f(w) = \lim_n f(f^n(z)) = \lim_n f^{n+1}(z) = w.$$

This completes the proof. ■